

SIROP: High-Level Synthesis with Parallel Patterns

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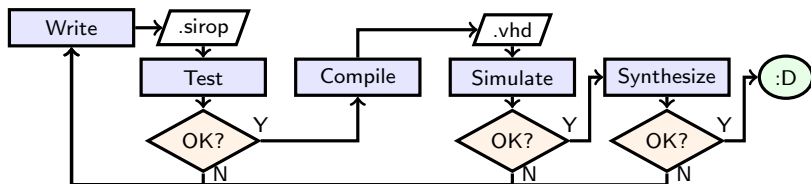


McGill

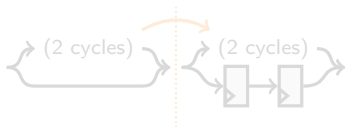
September 22, 2026

Recap: Project Goal

Goal: write high-level energy reconstruction algorithms, compile to resource-efficient VHDL



Latency matching



Fusion

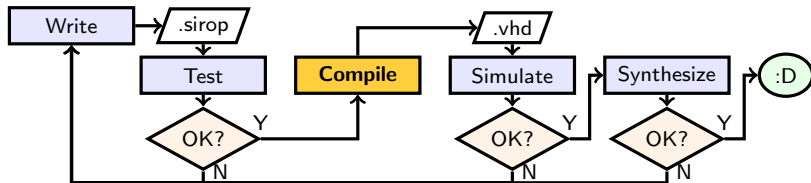


Fission

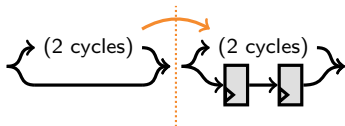
etc.

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Fusion

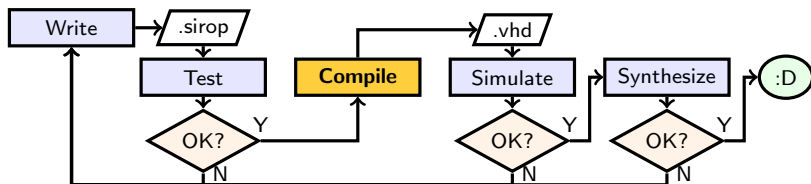
etc.



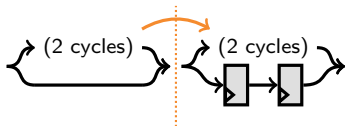
Fission

Recap: Project Goal

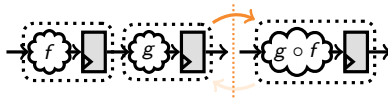
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Fusion

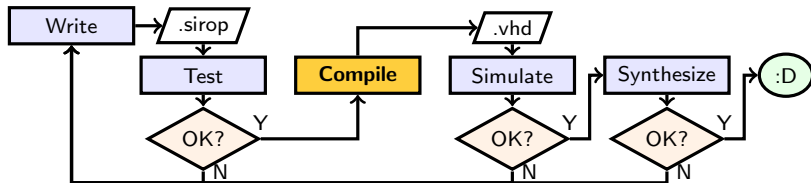


Fission

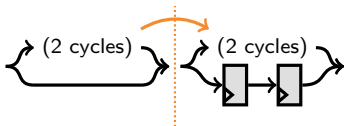
etc.

Recap: Project Goal

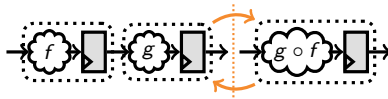
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Fusion



Fission

etc.

Previous Results, dacore FIR Filter

6-tap filter with hard-coded weights, targeting Agilex 7

Design	$f_{\max}$	ALMs	DSPs
Handwritten VHDL	880 MHz	123*	3
Generated w/ SIROP	510 MHz	98* (-15%)	3

*Excluding ALMs reported as “unavailable due to virtual I/Os.”

Why the Difference in ALM Usage?

1 Inconsistent compilation settings

- I didn't notice the device part number for McGill's devkit differs from the `G_BSP_AGI_NO_HW` target used in LASP-dacore :(
:(

2 Different DSP pipelining settings

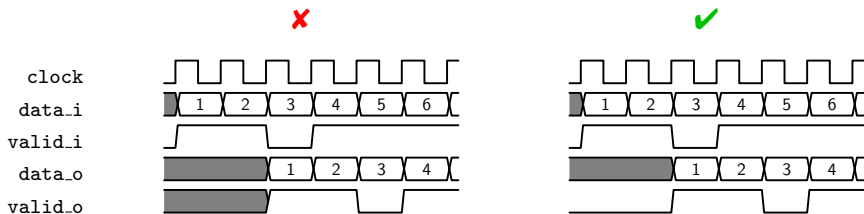
- See merge request [!143](#) in LASP-dacore
- See presentation explaining DSP register packing:
<https://indico.cern.ch/event/1700332/>

Updated Results, dacore FIR Filter

6-tap filter with hard-coded weights, targeting Agilex 7

Design	$f_{\max}$	ALMs	DSPs
Handwritten VHDL, before !143	880 MHz	123	3
Generated w/ SIROP	823 MHz	118	3
Handwritten VHDL, after !143	823 MHz	117	3

Initial valid Outputs



- It's good to ensure the valid field in `t_avst_stream` outputs is 0, even immediately after reset
 - See merge request [!142](#) in LASP-dacore
- SIROP can check such conditions (e.g., valid bit must be 0 for the first n cycles, where n is the entity's latency)

DSP Input Cascade

- TODO: explain FIR filter trick about selecting every 7th rather than every 8th index in shift register

SIROP can automatically transform the simpler version into this resource-efficient version (by merging and shrinking redundant shift registers)!

Conclusion

- Investigating differences between generated and handwritten VHDL led to ideas for improving resource-efficiency (!143) and robustness (!142)
- Even after merge requests, SIROP just about matches handwritten FIR filter, with much more concise source code
- Next steps: more advanced energy reconstruction algorithms
 - Neural networks (CPPM, Dresden)
 - Optimal filter with forward corrections (McGill)
- Future direction: faster exploration of neural net architectures (e.g., representation of weights, hypernetworks)



(SIROP
repository)